Hints: Functional and differential equations – 2. Functional-differential equations

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $f'(t) = f(\sqrt{t}) \geqslant 0$, so f is non-decreasing, and $f(x) = \int_0^x f(\sqrt{t})\,dt = \int_0^{\sqrt{x}} 2uf(u)\,du$. Aim: $f \equiv 0$ on $[0,1]$ first.

Hint 2. For $x \leqslant 1$, monotonicity gives $f(x) \leqslant xf(\sqrt{x}) \leqslant x \cdot x^{1/2} f(x^{1/4}) \leqslant \ldots$, hence $f(x) \leqslant x^2 f(1)$. Now feed a bound $f(u) \leqslant Cu^2$ into the integral: $f(x) \leqslant \int_0^{\sqrt{x}} 2Cu^3\,du = \frac{C}{2}x^2$. So C can be halved again and again: $f \equiv 0$ on $[0,1]$.

Hint 3. Let $s = \sup\{x : f = 0 \text{ on } [0,x]\} \geqslant 1$. If $s < \infty$, then for $x \in [s, s+\varepsilon]$ we have $|f(x)| \leqslant \varepsilon \max_{[s,s+\varepsilon]} |f(\sqrt{t})|$, and $\sqrt{t} \leqslant s + \varepsilon$; so $\max_{[s,s+\varepsilon]} |f| \leqslant \varepsilon \max_{[s,s+\varepsilon]} |f|$, which is 0 for $\varepsilon < 1$. Answer: $f \equiv 0$.

Problem 2

Hint 1. Condition (i) together with (ii) gives $f(a) = 0$ for every integer a. Look at f on a segment $[a, a+1]$.

Hint 2. If f took a positive value on $[a, a+1]$, its maximum there would be positive and attained at an interior point c, where $f'(c) = 0$; then (ii) gives $f(c) = 0$, a contradiction. The same for a negative minimum. Answer: $f \equiv 0$.

Problem 3

Hint 1. $f > 0$ on $[0, \infty)$: at the first zero $b > 0$ we would have $f'(b) = f(f(b)) = f(0) > 0$, while f decreases to 0 there. So $f' = f \circ f > 0$ on $[0, \infty)$, f increases there, and $f'(x) \geqslant f(f(0)) > 0$; hence $f(x) \to \infty$ and $f'(x) = f(f(x)) \to \infty$.

Hint 2. So $f(x) \geqslant x$ for $x \geqslant x_1$, and then $f'(x) = f(f(x)) \geqslant f(x)$, i.e. $f(x) \geqslant Ce^x$ for large x.

Hint 3. Therefore $f'(x) = f(f(x)) \geqslant Ce^{f(x)}$ for large x, i.e. $(e^{-f})' \leqslant -C$. A positive function cannot decrease with slope $\leqslant -C$ forever: contradiction. (In Differentiable functions – 1 the inequality $f' > f \circ f$ is excluded by the same growth argument.)

Problem 4

Hint 1. Regularity: $f'(x) = f(x+1)$ is differentiable, so $f \in C^2$, and so on: $f \in C^\infty$ and $f^{(n)}(x) = f(x+n)$.

Hint 2. Exponential solutions: $e^{\lambda x}$ works iff $\lambda = e^\lambda$. For real λ, $e^\lambda \geqslant 1 + \lambda > \lambda$, so all roots are non-real (they come in conjugate pairs, and give real solutions $e^{ax}\cos bx$, $e^{ax}\sin bx$).

Hint 3. All solutions: f is determined by $\varphi = f|_{[0,1]}$. On $[n, n+1]$, $f(x) = \varphi^{(n)}(x-n)$; on $[-1, 0]$, $f(x) = f(0) - \int_x^0 f(t+1)\,dt$, and so on to the left. Conversely, every C^∞ function φ on $[0,1]$ with $\varphi^{(n)}(1) = \varphi^{(n+1)}(0)$ for all $n \geqslant 0$ extends to a solution.

Problem 5

Hint 1. f is C^∞ and $f'(x) = \frac{f(x+h) - f(x-h)}{2h}$. Linear functions are solutions. An exponential $e^{\lambda x}$ is a solution iff $\frac{\sinh \lambda h}{\lambda h} = 1$, i.e. $z = \lambda h$ solves $\sinh z = z$.

Hint 2. The only real root is $z = 0$, but there are non-real ones: $\sinh z - z$ is an entire function of order 1, and if it had finitely many zeros it would be $P(z)e^{cz}$ (Hadamard), which contradicts its behaviour $\sim \frac{e^z}{2}$ as $z \to +\infty$ and $\sim -\frac{e^{-z}}{2}$ as $z \to -\infty$. So $e^{ax}\cos bx$ with $z = (a+ib)h$ is a non-linear solution.

Hint 3. Answer: all linear functions and many others (the closed span of such exponentials). If the mean value property holds for every $h > 0$, differentiate in h: $f(x+h) + f(x-h) = 2f(x)$, so $f'' = 0$ and f is linear — exactly as harmonic functions are the ones with the mean value property for all radii.

Problem 6

Hint 1. $x = y = 0$ gives $f(0) = -f'(0)^2$. If $f' \equiv 0$, then $f \equiv 0$. Otherwise pick y_0 with $f'(y_0) \neq 0$: $f'(x) = \frac{f(x+y_0)-f(x)-f(y_0)}{f'(y_0)}$, so $f \in C^\infty$.

Hint 2. Differentiate in y at $y = 0$: $f'(x)(1 - f''(0)) = f'(0)$. If $f''(0) \neq 1$, f' is constant and $f(x) = ax - a^2$.

Hint 3. If $f''(0) = 1$, then $f'(0) = 0 = f(0)$, and differentiating twice in y at 0 gives $f'' = 1 + kf'$. So $f' = x$ or $f' = \frac{e^{kx}-1}{k}$. Answer: $f(x) = ax - a^2$, $f(x) = \frac{x^2}{2}$, $f(x) = \frac{e^{kx}-1-kx}{k^2}$ ($k \neq 0$); for the last one $f(x+y) - f(x) - f(y) = \frac{(e^{kx}-1)(e^{ky}-1)}{k^2} = f'(x)f'(y)$.

Problem 7

Hint 1. Regularity: the harmonic mean of $2x$ and $\frac{2x}{3}$ is x, so $f'(x) = \frac{f(2x)-f(2x/3)}{4x/3}$ and $f \in C^\infty$.

Hint 2. General principle: put $a = x+h$, $b = x-h$, and write $\mu(x+h, x-h) = x + c(x)h^2 + O(h^3)$. Comparing the terms in h^3 gives $\frac{f'''}{3} = 2cf''$. For the arithmetic mean $c = 0$ (quadratics), for the geometric mean $c = -\frac{1}{2x}$ ($f'' = Kx^{-3}$, $f = Ax + \frac{B}{x} + C$), for the harmonic mean $c = -\frac{1}{x}$, so $f'' = Kx^{-6}$ and $f = \frac{A}{x^4} + Bx + C$.

Hint 3. This is only necessary. Linear functions satisfy the equation, so $\frac{A}{x^4}$ must satisfy it on its own; at $a = 2$, $b = 1$ it does not ($\frac{1}{16} - 1 \neq -4(\frac{3}{4})^5$), so $A = 0$. Answer: $f(x) = Bx + C$. The candidate from the ODE survives for the arithmetic and geometric means and dies for the harmonic one.

Problem 8

Hint 1. $f'(x) = f(1-x)$ gives $f \in C^2$ and $f''(x) = -f'(1-x) = -f(x)$, so $f = A\cos x + B\sin x$. Substituting back leaves a one-parameter family: $f(x) = C\cos(x - \frac{\pi}{4} - \frac{1}{2})$.

Hint 2. $f''(x) = f(-x)$ gives $f \in C^4$ and $f''''(x) = f''(-x) = f(x)$, so $f = ae^x + be^{-x} + c\cos x + d\sin x$. Substituting: $a = b$, $c = 0$. Answer: $f(x) = a\cosh x + d\sin x$.

Problem 9

Hint 1. $f(x) = f(\frac{x}{2^n}) \to f(0)$, so f is constant.

Hint 2. For $f'(x) = f(2x)$, $f(0) = 1$, the coefficients satisfy $(n+1)a_{n+1} = 2^n a_n$, so $a_n = \frac{2^{n(n-1)/2}}{n!}$.

Hint 3. $\frac{a_{n+1}}{a_n} = \frac{2^n}{n+1} \to \infty$: the radius of convergence is 0, so no solution is analytic at 0.

Problem 10

Hint 1. The characteristic equation is $\lambda = e^{-\lambda}$. It has no roots on the imaginary axis: $\lambda = ib$ would need $|b| = 1$ and $ib = \cos b - i\sin b$, i.e. $\cos b = 0$, which is false for $b = \pm 1$. The real root $\lambda_0 \approx 0.567$ grows at $+\infty$; all other roots have $\operatorname{Re}\lambda < 0$ and blow up at $-\infty$.

Hint 2. A clean proof uses the Fourier transform of the bounded function f (as a tempered distribution): the equation becomes $(i\xi - e^{-i\xi})\hat{f} = 0$. The factor $i\xi - e^{-i\xi}$ is smooth and never vanishes on $\mathbb{R}$, and $1/(i\xi - e^{-i\xi})$ is a smooth multiplier of polynomial growth; so $\hat{f} = 0$ and $f \equiv 0$.

Hint 3. So boundedness on $\mathbb{R}$ alone already forces $f \equiv 0$; the responsible property is that no characteristic root lies on the imaginary axis. The condition at $-\infty$ excludes the modes with $\operatorname{Re}\lambda < 0$ in an elementary approach.